

I BE EXAM MAY 2024
Electronics & Telecommunication
1AMRC1 Applied Mathematics-I

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Time-3 hrs

Maximum Marks [60]

Note-Attempt all the questions with any two of (a), (b) and (c) from each.

- Q.1 (a) Expand $\tan^{-1} x$ as Maclaurin's series and deduce that-

$$\tan^{-1} \frac{\sqrt{1+x^2}-1}{x} = \frac{1}{2} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right)$$

- (b) Find the radius of curvature at the point (r, θ) on $r^n = a^n \cos n\theta$.

- (c) Find all the asymptotes of the curve $(x+y)^2(x+2y+2) = x+9y-2$.

- Q.2 (a) If $u = f(x, y)$ and $x = r \cos \theta, y = r \sin \theta$ prove that-

$$\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2$$

- (b) Find the possible percentage error in computing the resistance r from the formula $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$ if r_1, r_2 are both in error by 2%.

- (c) Use Lagrange's method of undetermined multipliers to divide 24 into three parts such that the continued product of the first, square of the second and the cube of the third may be maximum.

- Q.3 (a) Show that $\beta(m, n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$. Hence evaluate $\int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}} dx$.

- (b) Find the area of the loop of the curve $x^3 + y^3 = 3axy$.

- (c) Show that the whole length of the curve $8a^2y^2 = x^2(a^2 - x^2)$ is $\pi a\sqrt{2}$.

- Q.4 (a) Find the area included between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ by double integration.

- (b) Evaluate $\int_A x^2 dx dy$, where A is the region in the first quadrant bounded by hyperbola $xy = 16$ and the lines $y = x, y = 0$ and $x = 8$.

- (c) Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy dx dy$ and hence evaluate it.

- Q.5 (a) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$.

- (b) Evaluate $\int \vec{F} \cdot \hat{n} dS$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of cube bounded by the plane $x = 0, x = a, y = 0, y = a, z = 0, z = a$.

- (c) State Stoke's theorem and evaluate $\int [(2x-y)dx - yz^2dy - y^2zdz]$ where C is the circle $x^2 + y^2 = 1$, corresponding to the surface of sphere of unit radius.

$$y = \sqrt[3]{3axy - x^3}$$

$$2\sqrt{2}(-2a+a)$$